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Student performance prediction



Introduction

Education is essential for leading a productive life and accessing necessary resources. As technology, including artificial intelligence, becomes increasingly prevalent, education institutions are leveraging these tools to improve students' performance. There is a growing focus on predicting academic success, as strong academic performance enhances an institution's ranking and boosts students' job prospects in the future. Grade prediction involves analyzing various factors to estimate a student's future academic performance. This process typically considers historical data, such as past grades, attendance records, and participation in class. Additionally, external influences like socioeconomic status, parental involvement, and access to resources can significantly impact a student's likelihood of success.

Problem statement

In Portugal, secondary education consists of 3 years of schooling and most of the students join the public and free education system. Despite the remarkable evolution in the Portuguese population's level of education over the past decade, the country's school results still leave it lagging behind the rest of Europe. These failures are mainly critical in the basic courses of mathematics and Portuguese. Decision-makers want to analyze the factors influencing student performance and develop a strategic plan to improve results. This study analyzes student achievement in secondary education across two Portuguese schools. We aim to investigate the potential for Portuguese government decision-makers to improve educational outcomes.

Dataset

Source: [Student Performance - UCI Machine Learning Repository](#)

The data encompasses various attributes, including student grades, demographic factors, and social and school-related characteristics, collected through school reports and questionnaires.

Two datasets are included, reflecting performance in two specific subjects:

- Mathematics: 395 rows and 34 columns
- Portuguese language: 649 rows and 34 columns

There are 34 attributes in the two datasets, they can be categorized as follows:

- Demographic and social: age, sex, address, family structure, etc
- Historical academic data: absences, failures, past grades, etc
- Target variable: Grade of the third year in second school (G3)

Categorical attributes:

- Binary categorical values with only two possible values: *sex, school, adress, family size, parent's cohabitation status, school support, family support, paid*

support, extra-curricular activities, attended nursery school, wants to take higher education, Internet access at home, with a romantic relationship.

- Nominal categorical values : *mother job, father job, reason to choose this school, student's guardian.*
- Ordinal categorical values: *mother's education, father's education, home to school travel time, weekly study time, quality of family relationships, free time after school, going out with friends, workday alcohol consumption, weekend alcohol consumption, current health status.*

Numerical attributes : age, number of failures, number of absences, and the three grades in second school (G1,G2, and G3)

Data wrangling

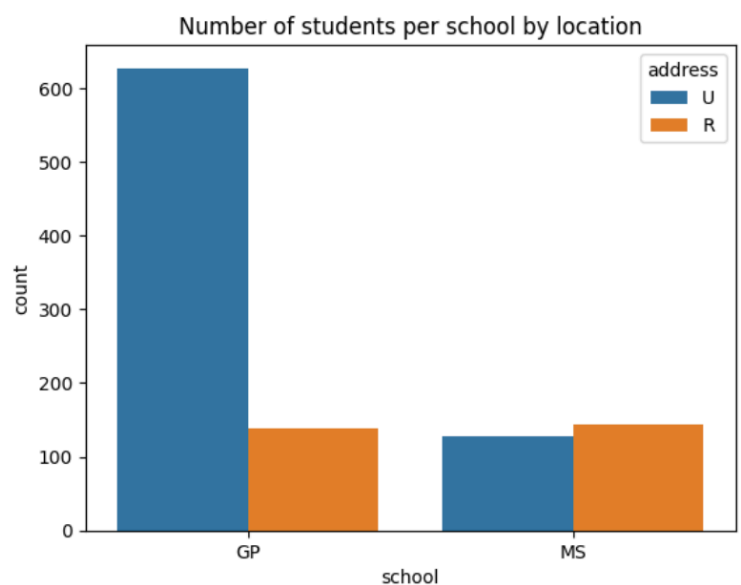
We First concatenate the two DataFrames and add a new column course. No missing or redundant data was present in the dataset. The ordinal categorical variables are already encoded in the original dataset. To obtain a full numerical dataset, we performed the following operations:

- a one-hot encoding on the nominal categorical variables and 5 of the binary categorical variables namely *sex, school, address, family size, and parent's cohabitation status.*
- 0/1 mapping of the yes/no values of the rest of the binary values

Exploratory data analysis

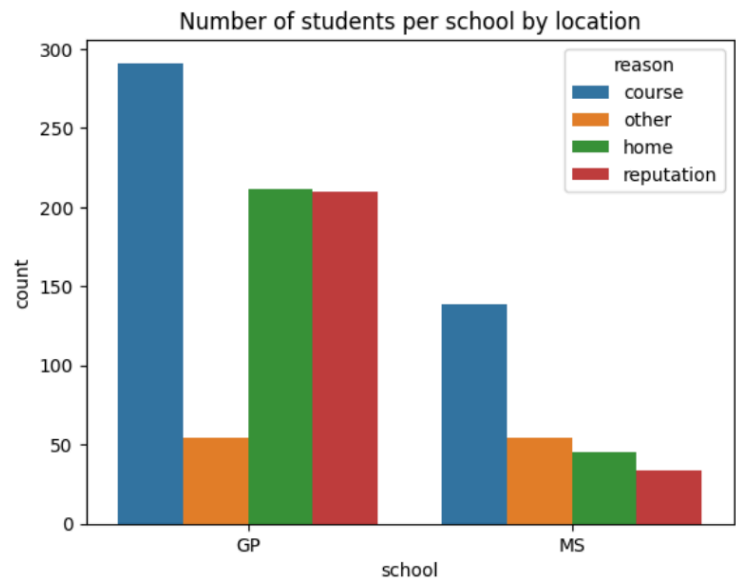
School differences

We first want to visualize the distribution of students in the two schools regarding their location (urban or rural). The GP school has more students than the MS

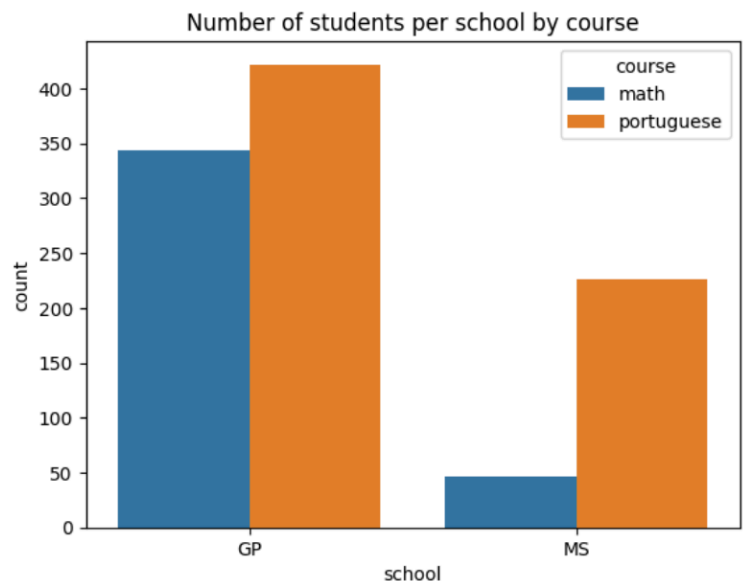


school. Students at the GP School come mainly from urban areas, while the MS School enrolls a balanced proportion of students from urban and rural areas.

Students' choice of school was mainly based on the courses offered by the two schools. A good proportion of students also chose the GP school because of its proximity to their home and reputation.

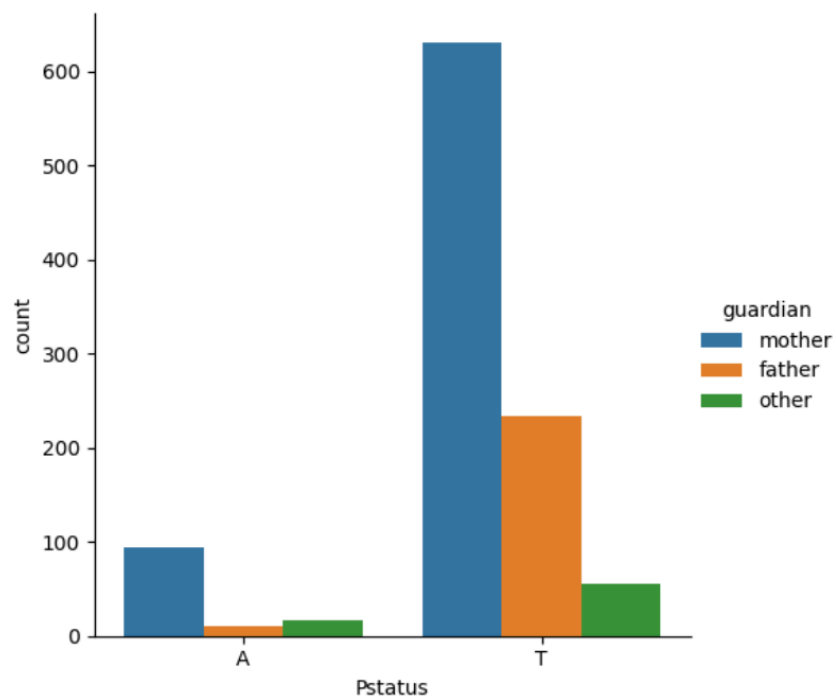


The distribution of students in the two schools regarding Math and Portuguese courses shows that most of the data relates to the Portuguese course at the two schools.

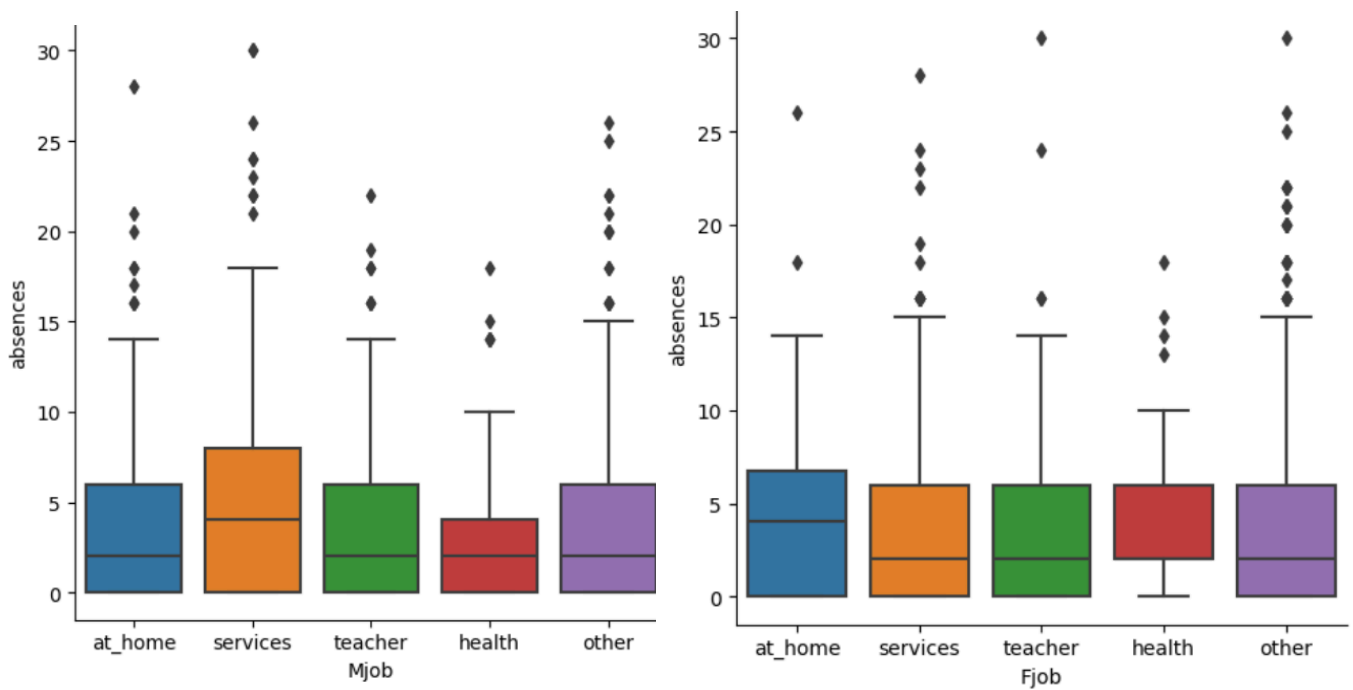


Students' environment

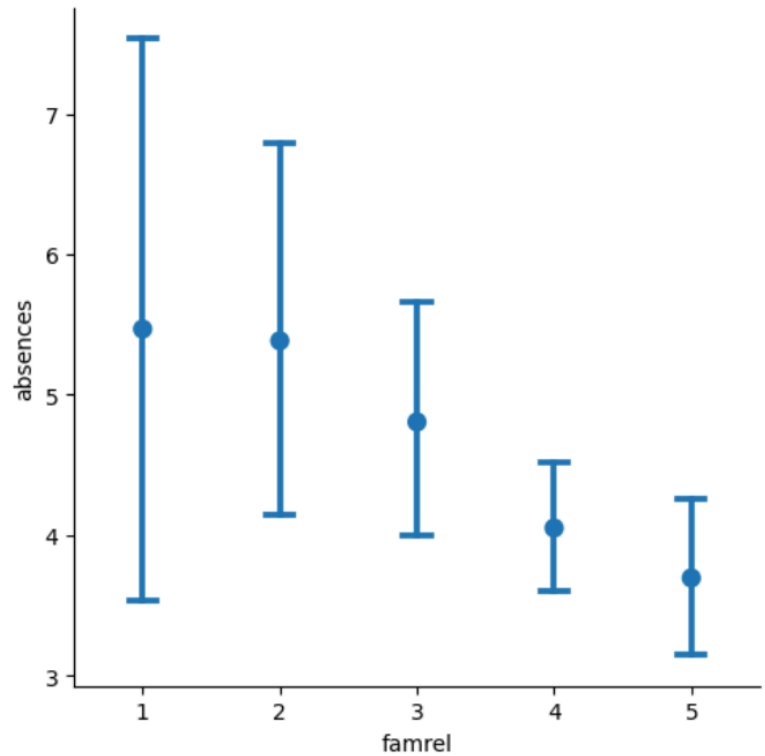
There are fewer students from separated families in the data and most students are guarded by their mothers, regardless of whether their parents are separated or together.



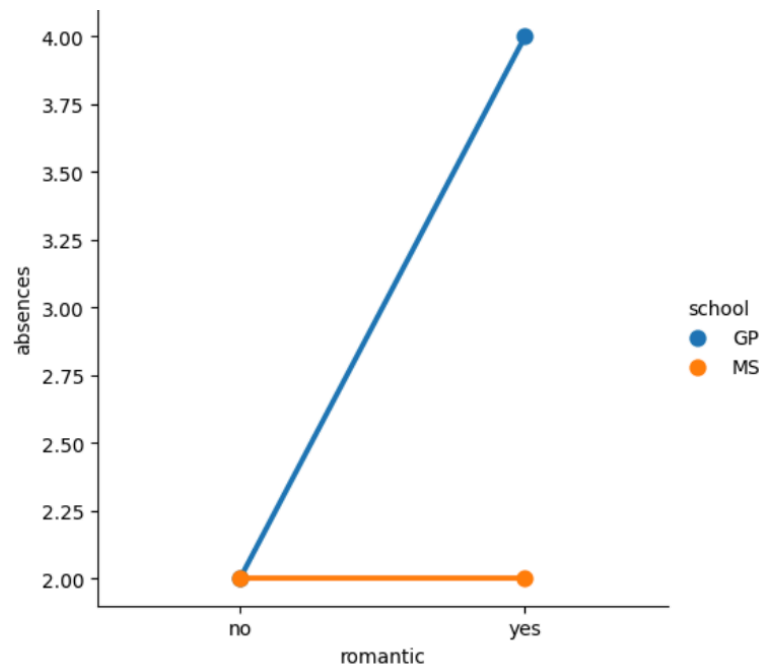
The mother's job has a more effect on student absences than the father's. This is to be expected since the mother is the main guardian.



Although the average number of absences is slightly lower among students with better family relationships, the wide confidence intervals indicate that we cannot be sure that there is a true association.

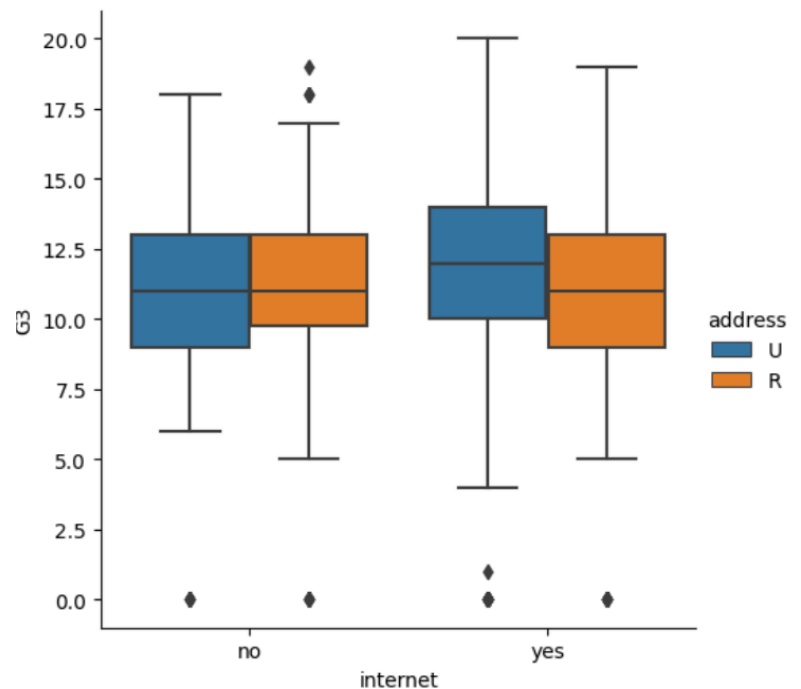


Students in a romantic relationship have a higher median number of absences in GP school, but this association does not hold true in MS school.



Students' grades

Internet access brings no benefit to students from rural areas, their grades have not been affected by the availability of the internet and, in some cases, it has even worsened their school results, perhaps due to improper use. A slight improvement is observed in the results of students from urban areas.



We also notice that:

- Students studying more time are less likely to skip classes and have better grades regardless of the course.
- There is a correlation between G1 and G3 regardless of the type of support received by the student.
- The median and maximum grades are higher among students who are not in a romantic relationship. Female students' grades are less affected by these relationships.

Features relationship

To assess whether the means of the target variable G3 differ significantly across different categories of categorical variables, we performed ANOVA (Analysis of Variance). We found that the means of the target variable G3 are not significantly

different across the levels of the categorical variables sex, family size, parent's status, family support, paid support, activities, nursery, and family relationship.

We computed the correlation between numeric features and we found that there is a strong positive correlation between the three grades exceeding 0.8. We therefore decided to abandon the two previous grade columns and add another column representing the change in a student's grades, i.e. the difference between the second and first year.

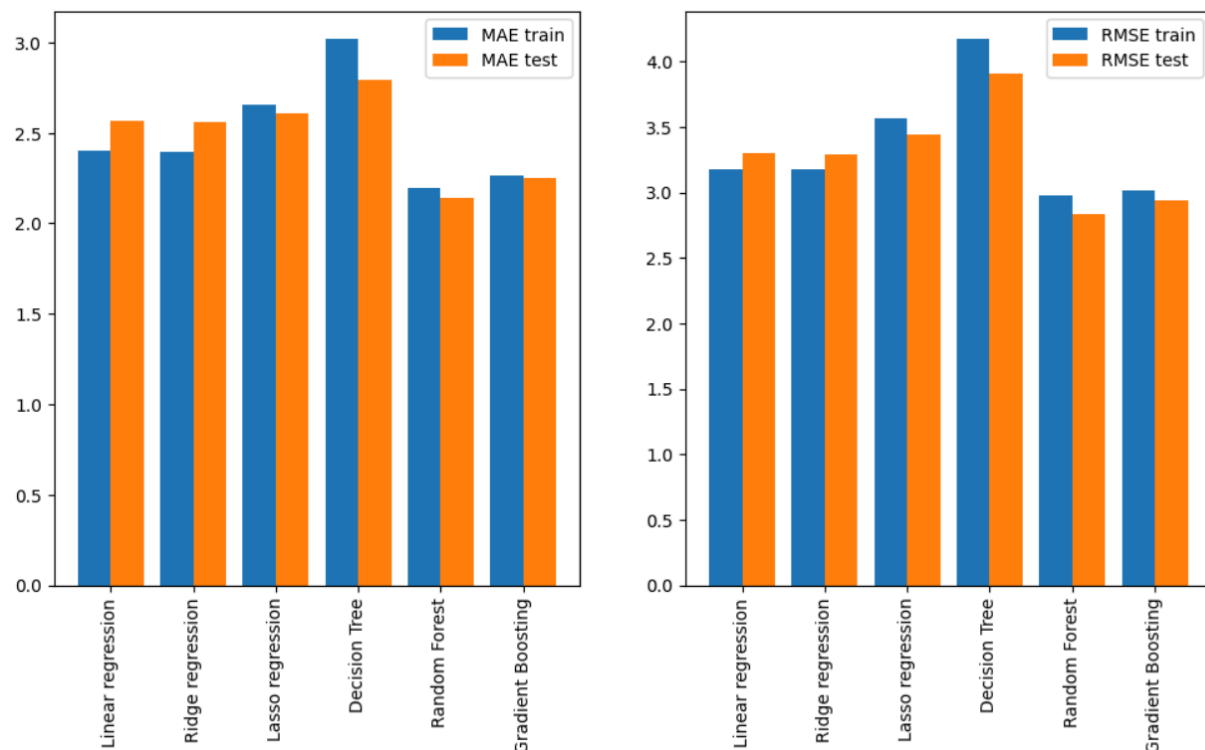
Modeling

we trained 6 models:

1. Linear regression
2. Ridge regression
3. Lasso regression
4. Decision Tree
5. Random Forest
6. Gradient boosting regressor

For each model, the mean absolute error (MAE) and the root mean squared error (RMSE) are calculated. For the linear regression, ridge and lasso models, we used scaled data.

Models evaluation showed that decision tree model gives the worst results in terms of mean absolute error and root mean squared error followed by Lasso. Linear regression and ridge regression give similar results, the regularization introduced in ridge has minor effect in prediction. The best two models are Random Forest and gradient Boosting.



We next tuned the hyperparameters of Random Forest and Gradient Boosting:

- Best parameters of Random Forest:
`{'criterion': 'squared_error', 'max_depth': 20, 'n_estimators': 100}`
- Best parameters of Gradient Boosting:
`{'learning_rate': 0.25, 'loss': 'squared_error', 'max_depth': 3, 'n_estimators': 50}`

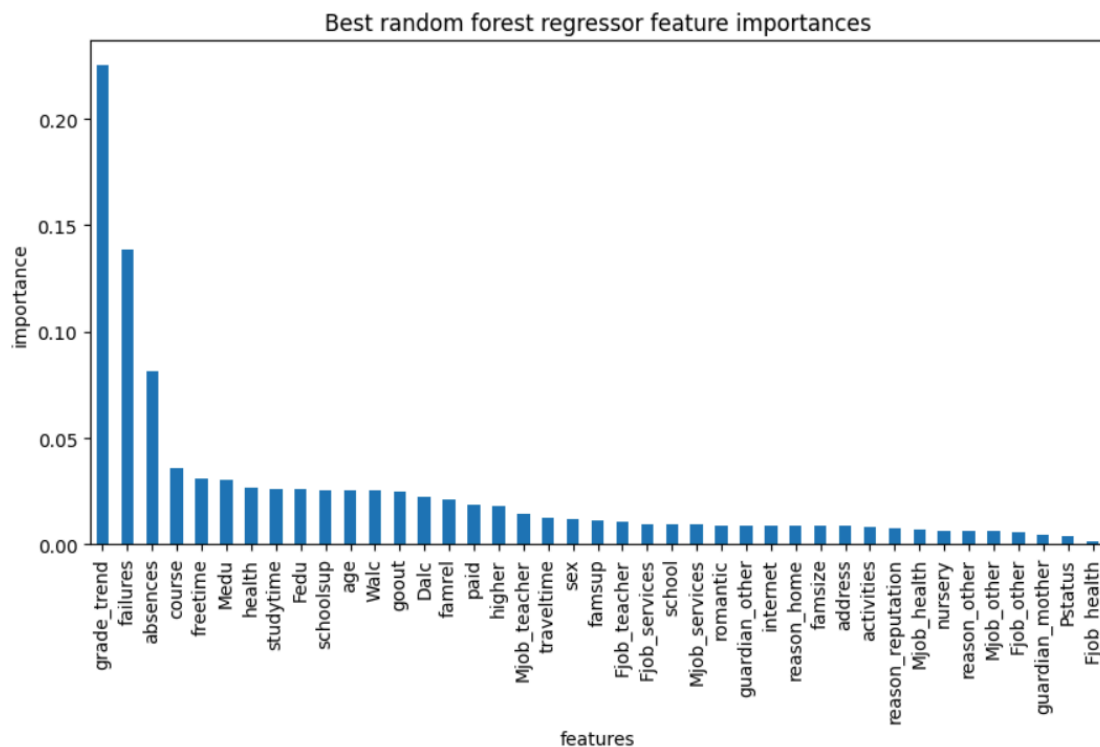
After tuning the hyperparameters for the best two models, we obtained the following results:

Model	Mean MAE train	Std MAE train mean	MAE test	Mean RMSE train	Std RMSE train	RMSE test
Random Forest	2.196709	0.182938	2.140584	2.978908	0.254237	2.833533
Gradient Boosting	2.252657	0.082890	2.264966	2.993485	0.119121	2.942300

The Best fitted model is Random Forest with a test MAE of 2.14 and a test RMSE of 2.83.

Features impact

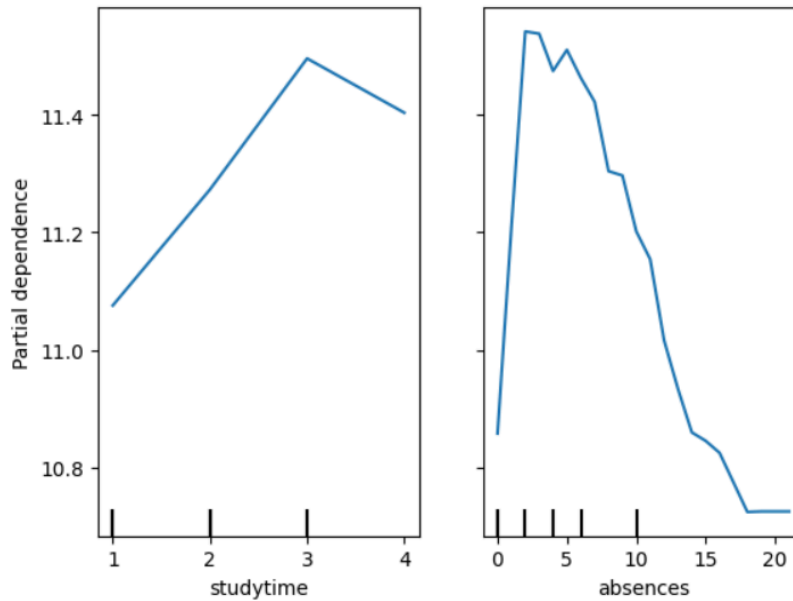
For Random Forest, the most important feature is grade_trend which represents the difference between the first and the second year of study, the impact of the other features is given below.



If the Portuguese government can put in place a framework to reduce school failure by 80% and deteriorating grades, the success rate can be increased by about 11%.

It is hard to control some of the most important features directly. We propose to monitor the number of absences and study time and observe the impact on grades prediction using the Random Forest model.

To visualize the relationship between these two features and the predicted outcome of a model, we used partial dependence plots (PDPs).



Our recommendations using these two features include

1. Imposing severely restrictive limits on the number of absences (No more than 3 per course).
2. Encourage students to spend 5 to 10 hours per week on their studies (homework, projects,...).

This scenario allows enhancing the success rate to above 7%.

Conclusion

In this project, we are interested in predicting student performance. The dataset provides information on the student's environment and academic performance, and we aim to fit a model capable of accurately predicting the students' final grades. In our experimentations, we tested six models and we selected the best one based on mean absolute error and root mean squared error. Random Forest gives the lowest errors, after parameters tuning, the best Random Forest fitted model allows predicting the final grade with an error of 2.14. We additionally studied the

impact of the number of absences and study time on grades. By limiting the number of absences to 3 and ensuring a weekly study time of 5 to 10 hours, it is possible to improve the success rate by 7%. In future work, we can explore automatic feature selection methods to remove some irrelevant features.